



BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY

**FACULTY OF COMPUTING DEPARTMENT OF
SOFTWARE ENGINEERING**

OSSP PROJECT – INDIVIDUAL ASSIGNMENT

Operating System: BackBox Linux

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1. Installation of BackBox Linux in a Virtual Environment

a. Introduction

BackBox Linux is an Ubuntu-based Linux distribution developed for penetration testing, security auditing, and system analysis. It provides a wide range of tools for vulnerability assessment, digital forensics, exploitation, and network security testing.

BackBox Linux is lightweight, fast, and highly customizable. It uses the XFCE desktop environment, which ensures efficient performance, especially when deployed in a virtual environment. In this project, BackBox Linux is installed inside a virtual machine to explore its features, evaluate its filesystem support, and understand the concept and practical application of virtualization in operating system installation.

b. Objectives

The main objectives of this project are:

- To install BackBox Linux on a virtual machine using VirtualBox or VMware Workstation.
- To explore the operating system environment and understand virtualization concepts.
- To examine the filesystem support provided by BackBox Linux.
- To document the installation process, challenges encountered, and the corresponding solutions.

c. Requirements

i. Hardware Requirements

The minimum hardware requirements for installing BackBox Linux in a virtual environment are:

- **Processor:** Dual-core processor or better
- **RAM:** Minimum 2 GB (4 GB recommended)
- **Storage:** At least 20 GB of free disk space

For this project, the host machine used runs on **Windows 11** and is equipped with **16 GB RAM** and approximately **350 GB of available storage**, which exceeds the recommended requirements. This ensures smooth and efficient performance of the virtual machine.

ii. Software Requirements

- VMware Workstation or Oracle VM VirtualBox
- BackBox Linux ISO image

- A host operating system such as Windows, Linux, or macOS

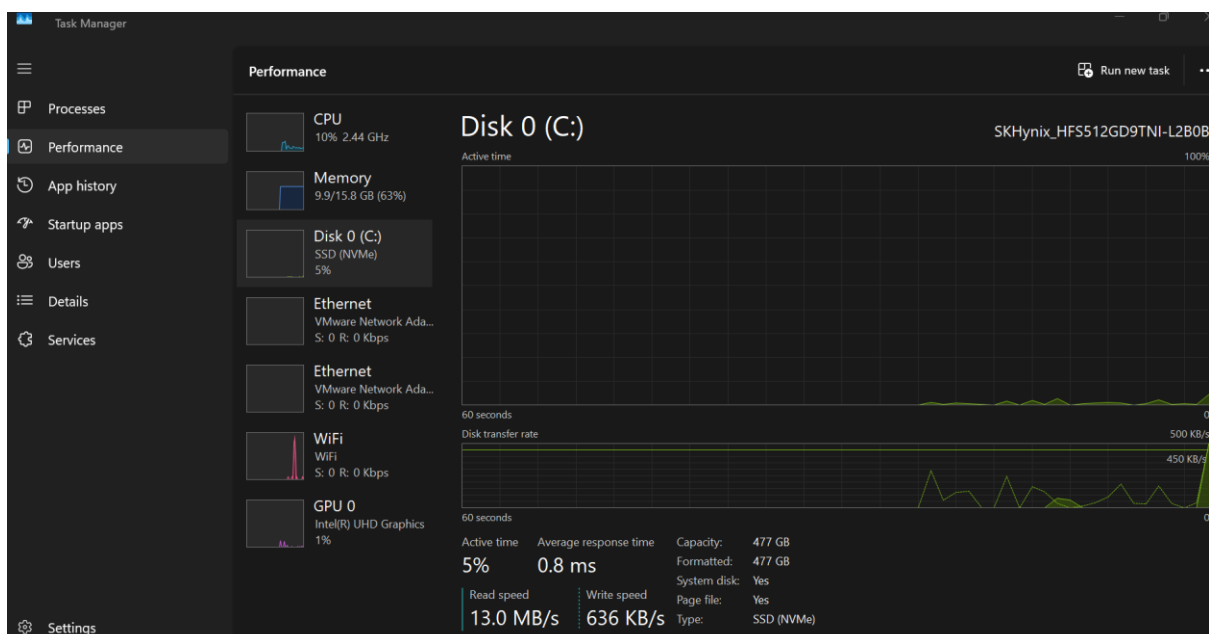
The host operating system used for this project is **Windows 11**.

d. Installation Steps

Step 1: Verify System Requirements

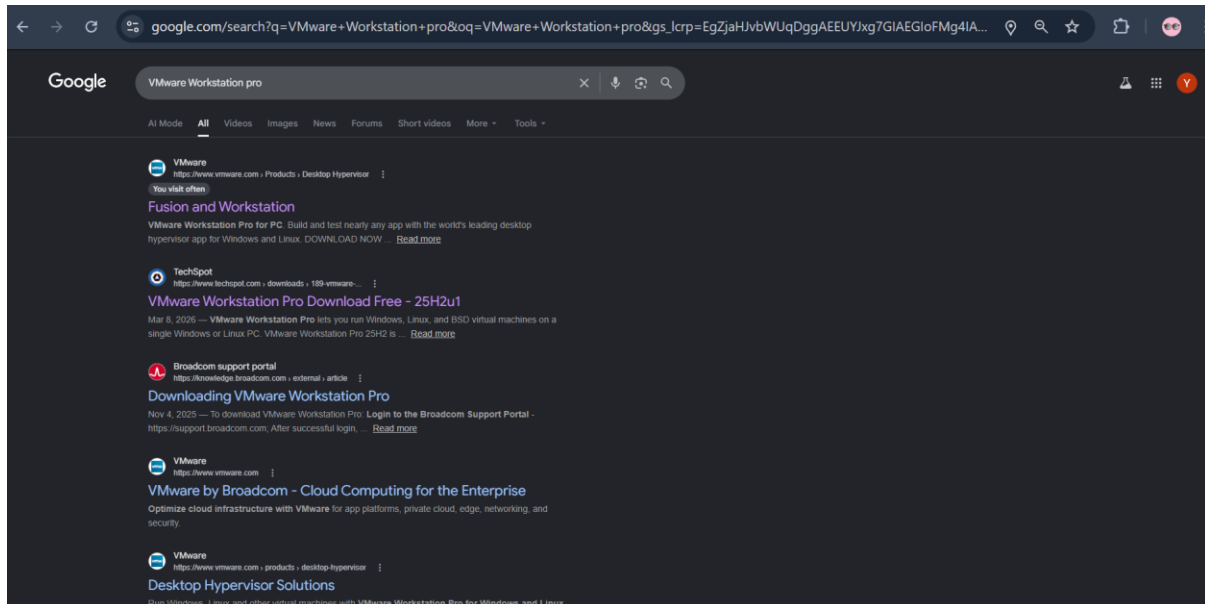
First, ensure that the computer meets the minimum requirements necessary to install BackBox Linux. This includes checking the processor, RAM, and available storage to confirm compatibility with the virtual environment.

On a Windows system, this can be done by opening **Task Manager** using the shortcut **Ctrl + Shift + Esc**. Then navigate to the **Performance** tab to view system details such as CPU, memory (RAM), and disk usage

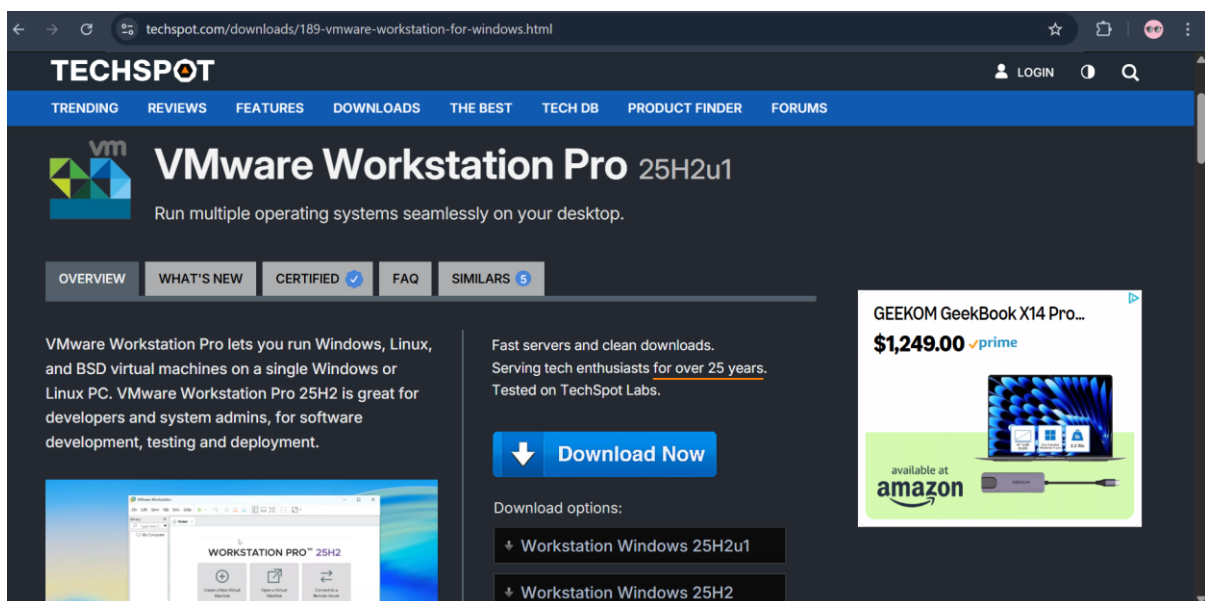


Step 2: Download VMware Workstation pro

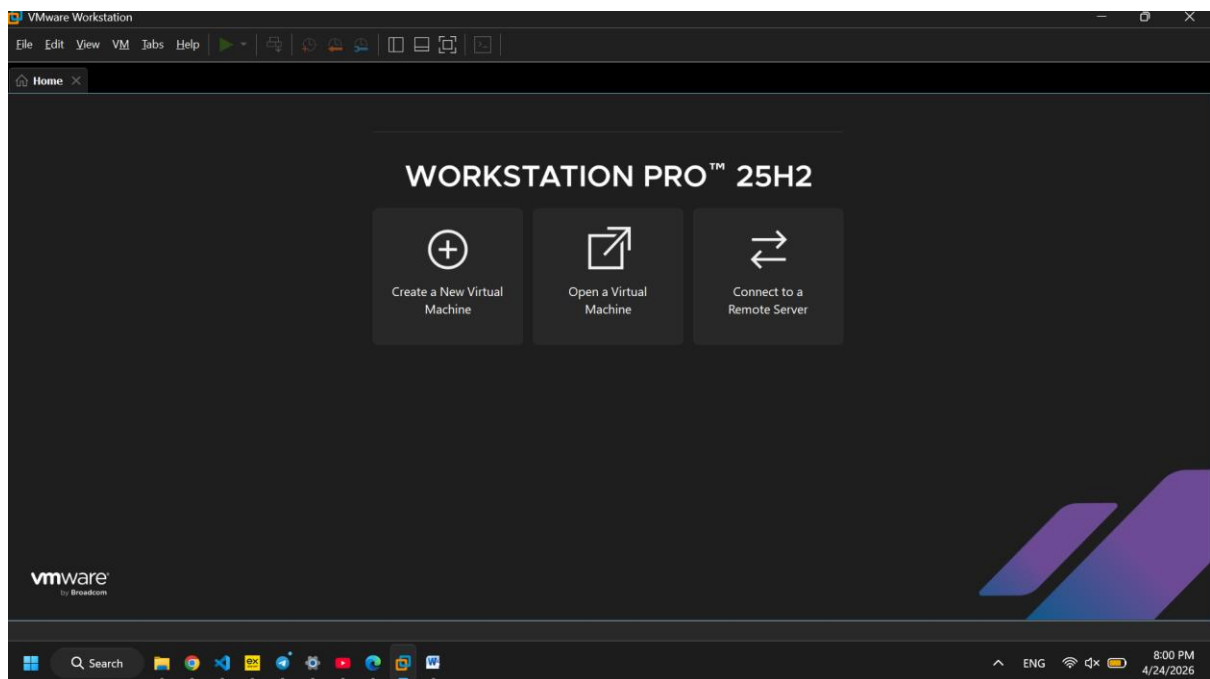
Open a web browser and search for “**VMware Workstation pro**”. Open a web browser and type “**VMware Workstation**” in the search bar. From the search results, select a reliable source such as TechSpot or the official VMware website.



After opening the download page, choose the appropriate version based on your operating system (Windows or macOS), then proceed to download the installation file

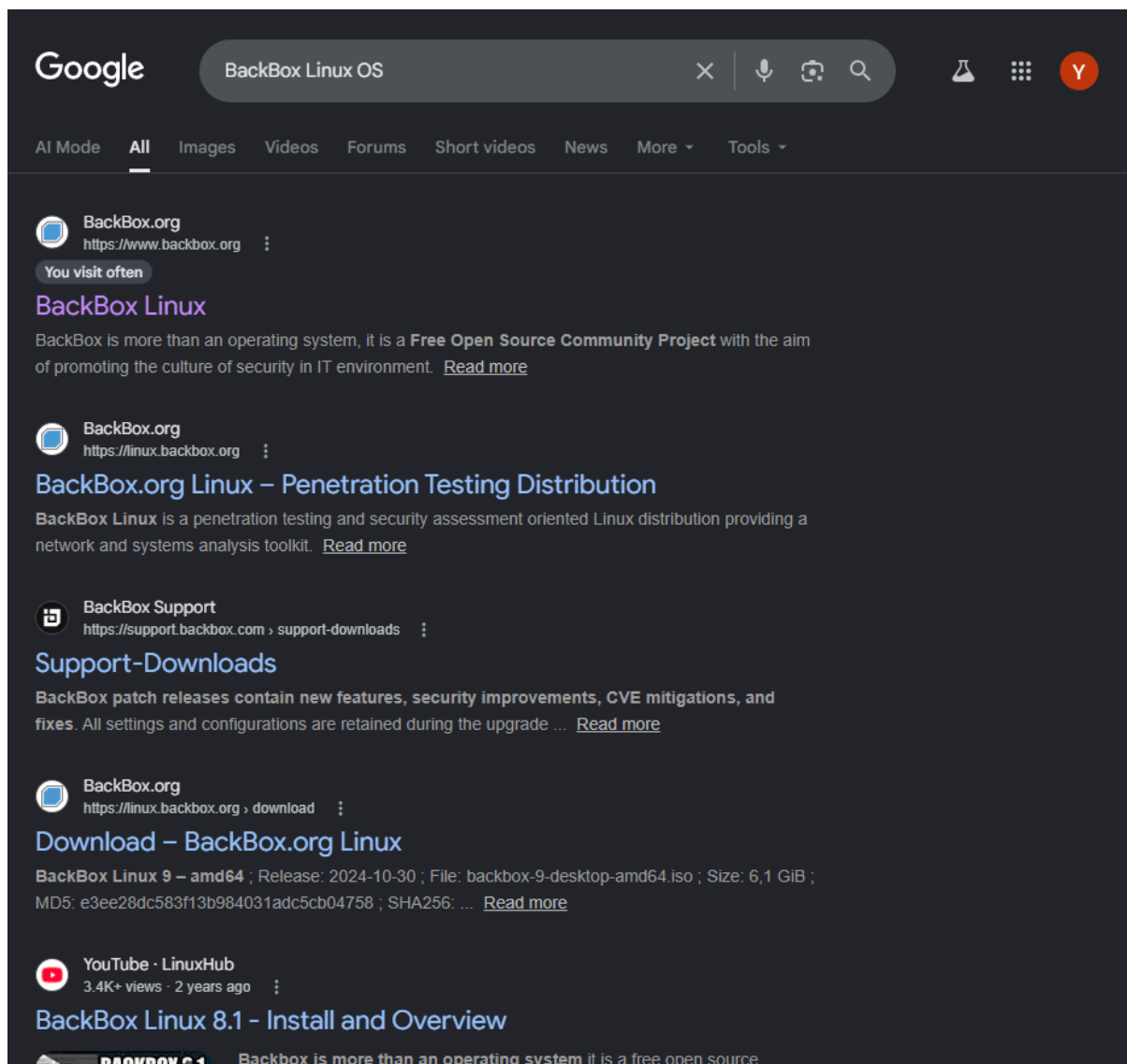


Once the download is complete, open the installer and follow the on-screen instructions to install VMware Workstation. The installation process includes accepting the license agreement, selecting installation options, and completing the setup.

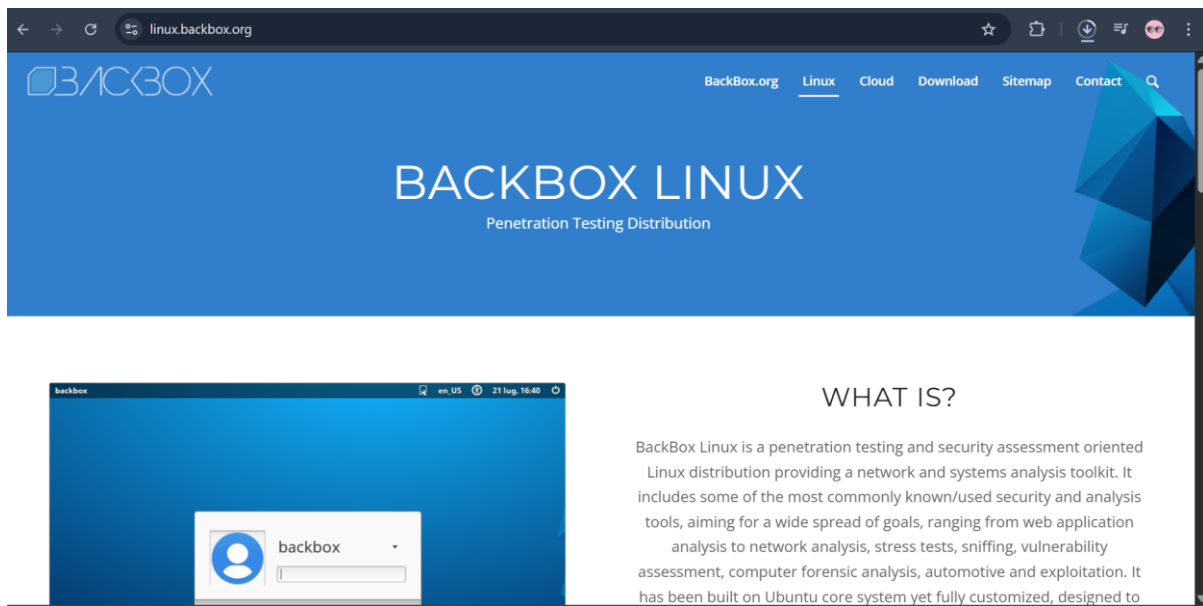


Step 3: Download BackBox Linux OS

Open a web browser and type “**BackBox Linux OS**” in the search bar. From the search results, visit the official website or the first trusted source that appears.



On the official website, locate the download section and select the appropriate **BackBox Linux ISO image** for download.

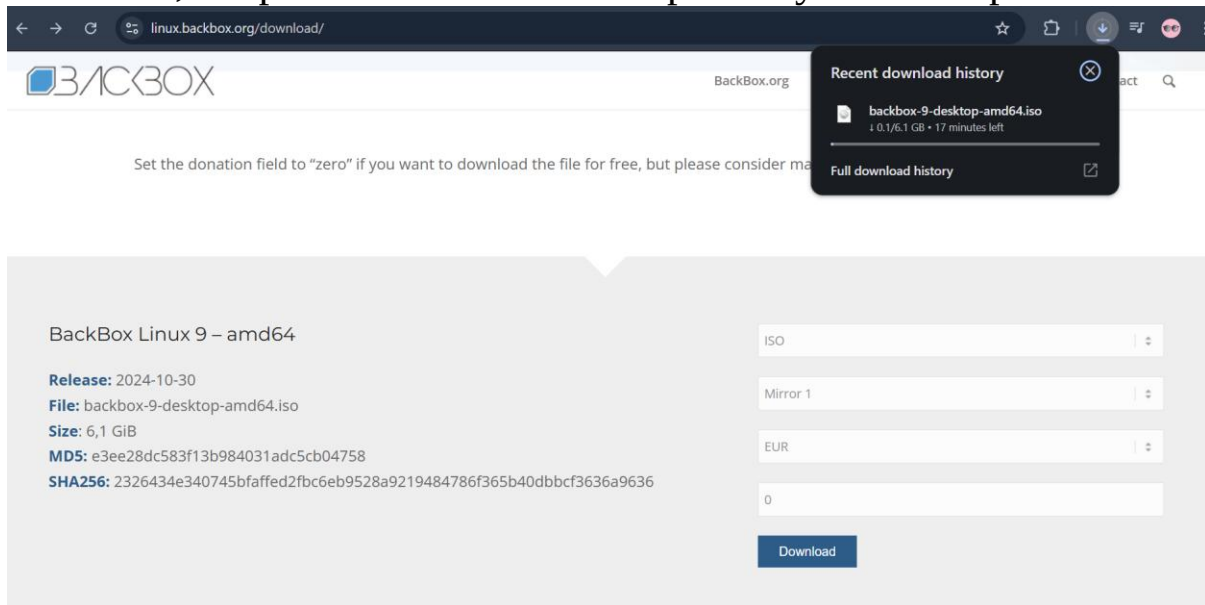


Important: Make sure to select and download the **ISO file only**, not the **torrent file**, because the ISO file is required for installation in VMware or VirtualBox.

After selecting the correct ISO file, the download process will begin. Once completed, the BackBox Linux ISO image will be available on the system and ready to use for virtual machine installation.

Since the ISO file size is relatively large, the download may take some time depending on the internet speed and system performance.

Therefore, the process should be done patiently until completion

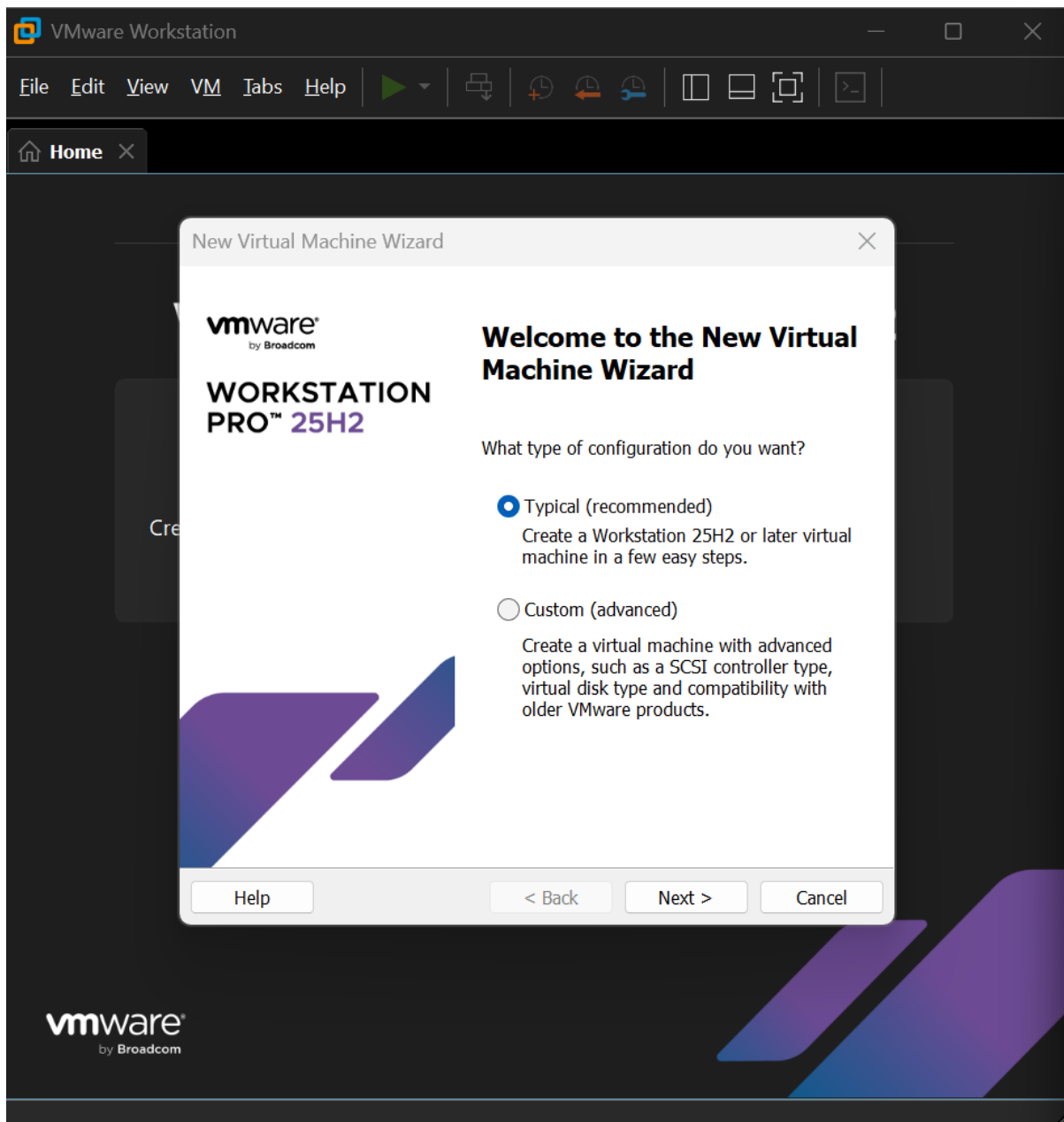


Step 4: Install BackBox Linux on VMware Workstation 17

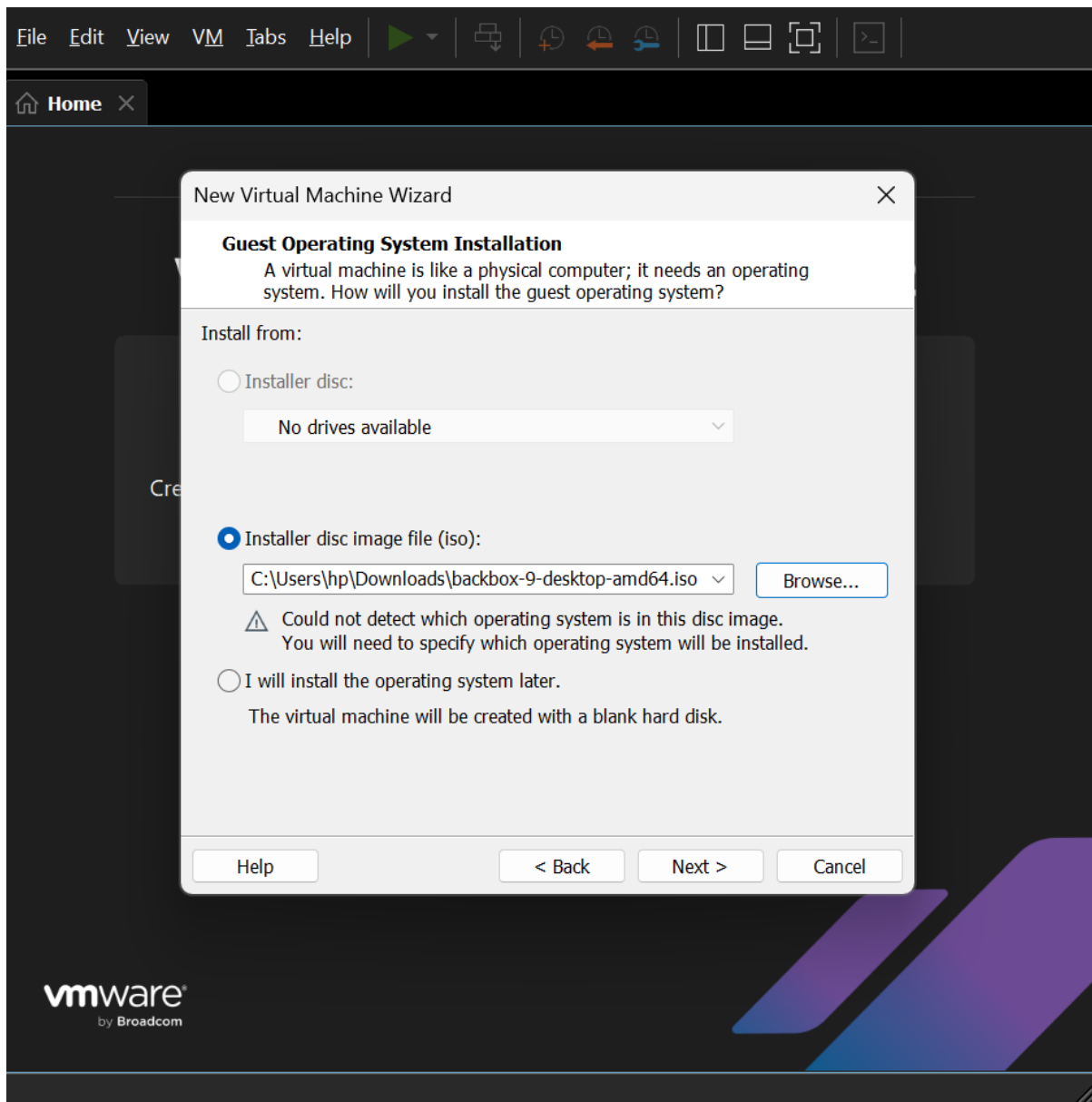
Open **VMware Workstation 17** and click on “**Create a New Virtual Machine**”.

Follow the installation wizard step by step as shown in the figures below:

1. Select **Typical (Recommended)** and click **Next**

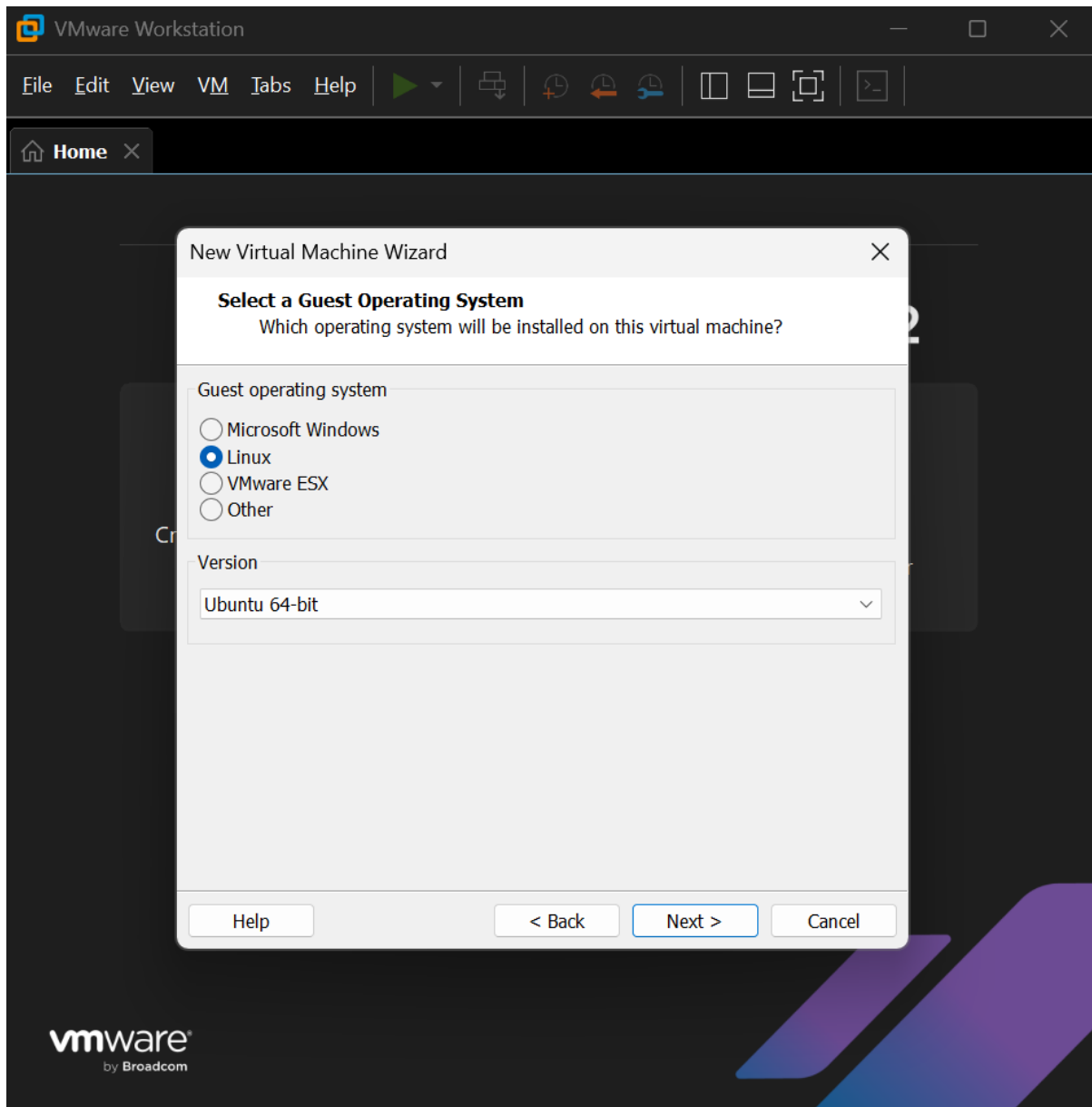


2. Choose **“Installer disc image file (ISO)”**, then click **Browse** and select the downloaded BackBox Linux ISO file.

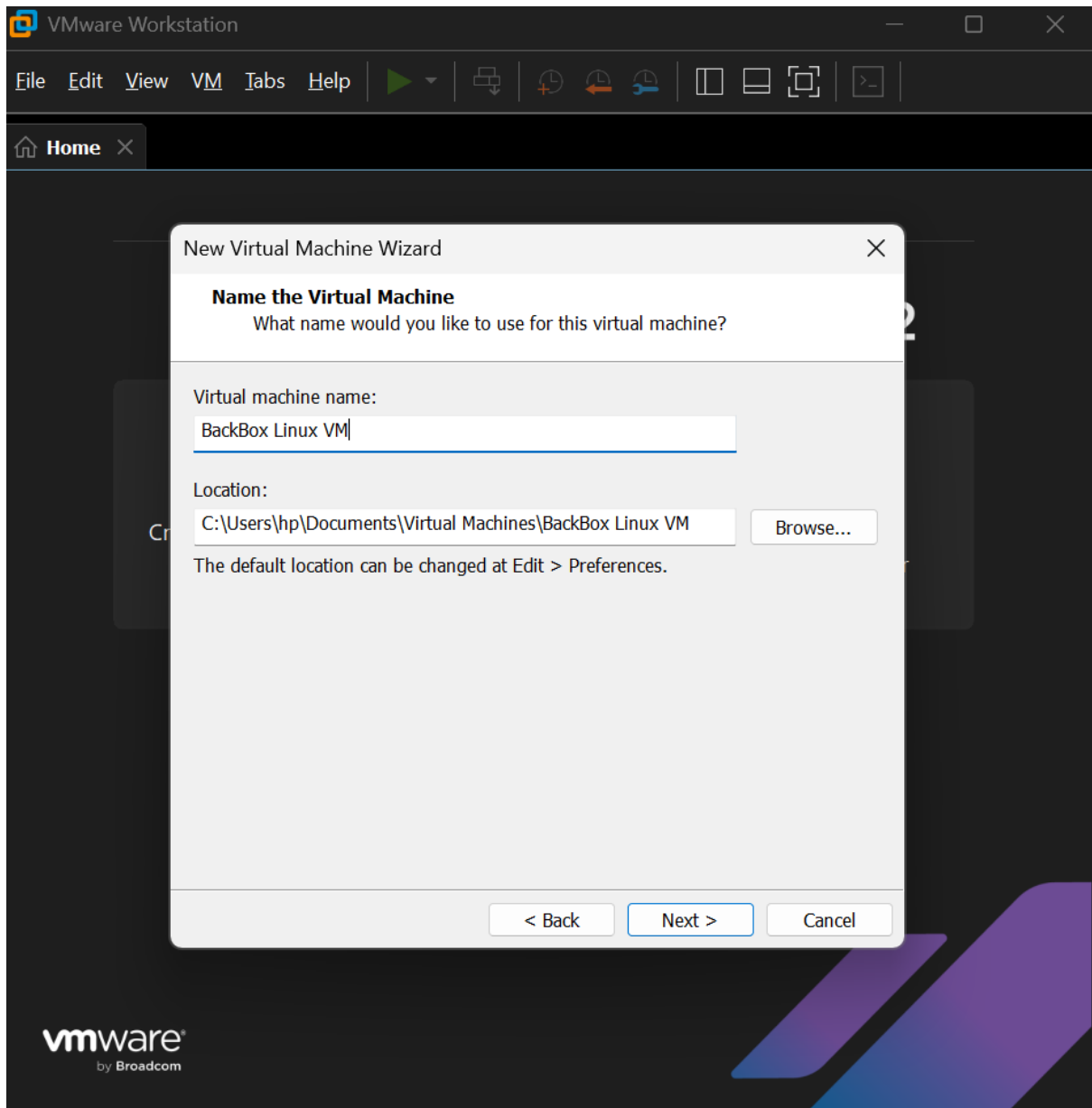


3. Select the guest operating system:

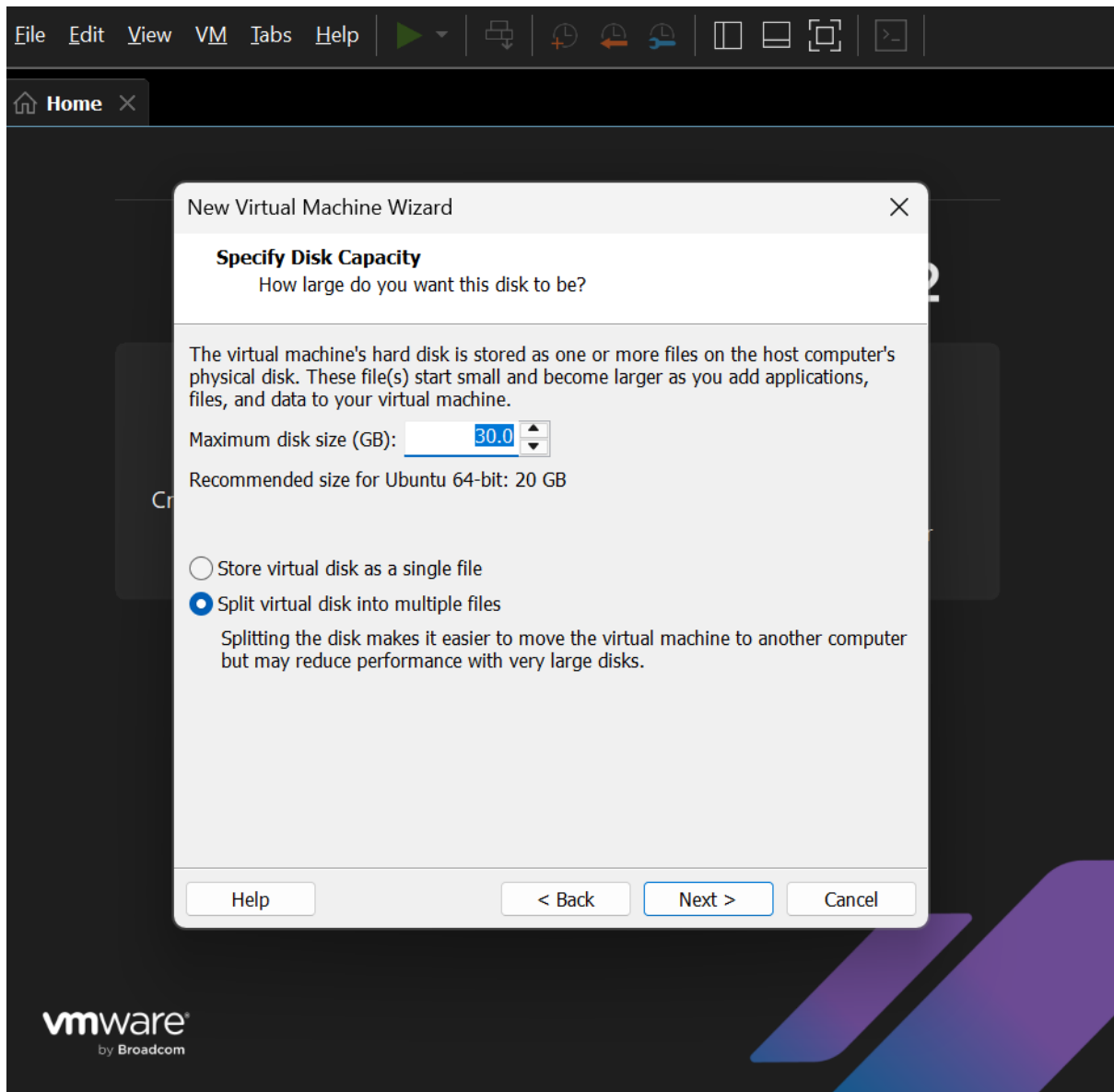
- **Operating System:** Linux
- **Version:** Ubuntu 64-bit



4. Enter the virtual machine name (e.g., **BackBox Linux 9**) and choose the storage location.



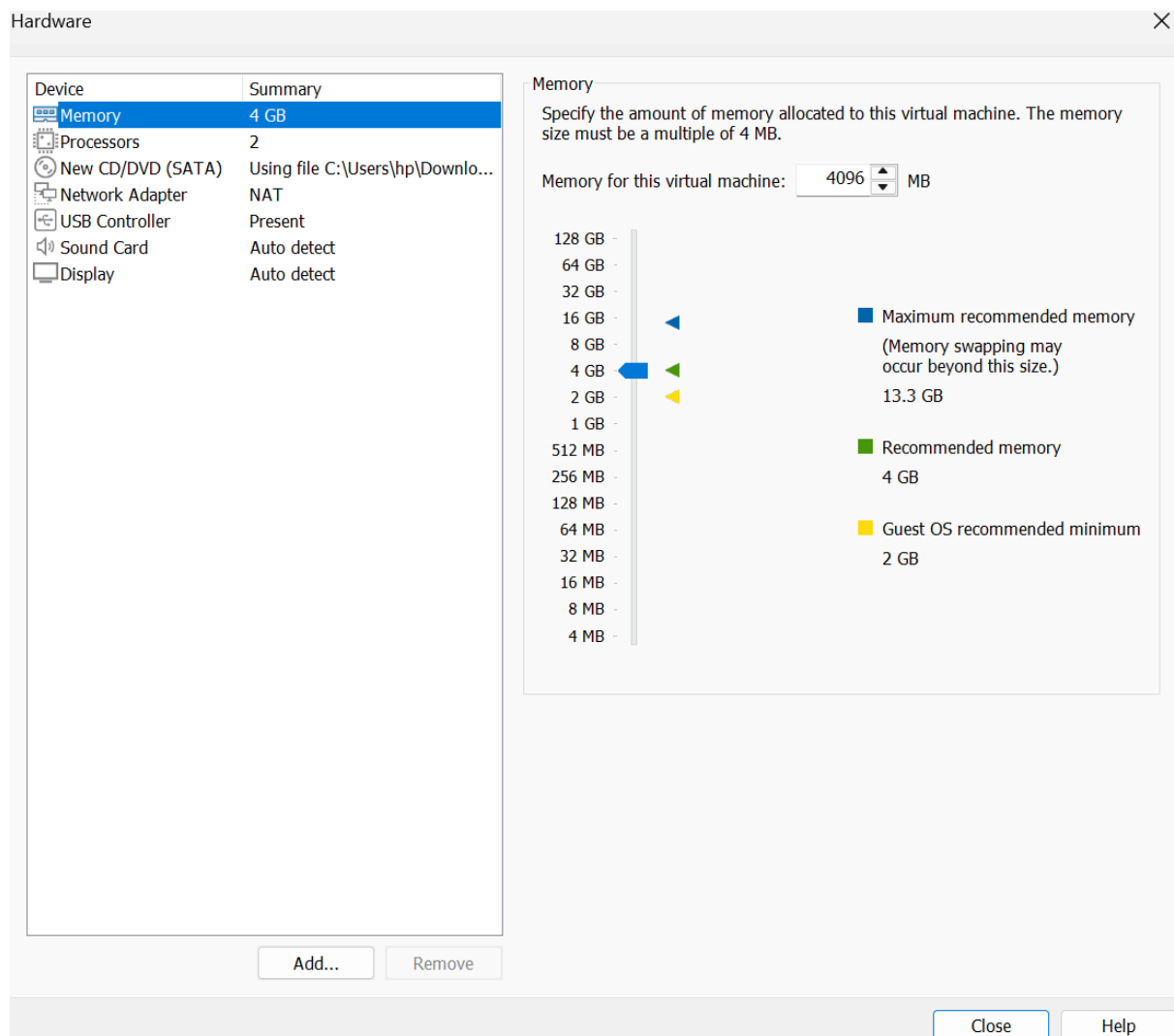
5. Set the disk size (recommended 30 GB) and select **“Store virtual disk as a single file.”**



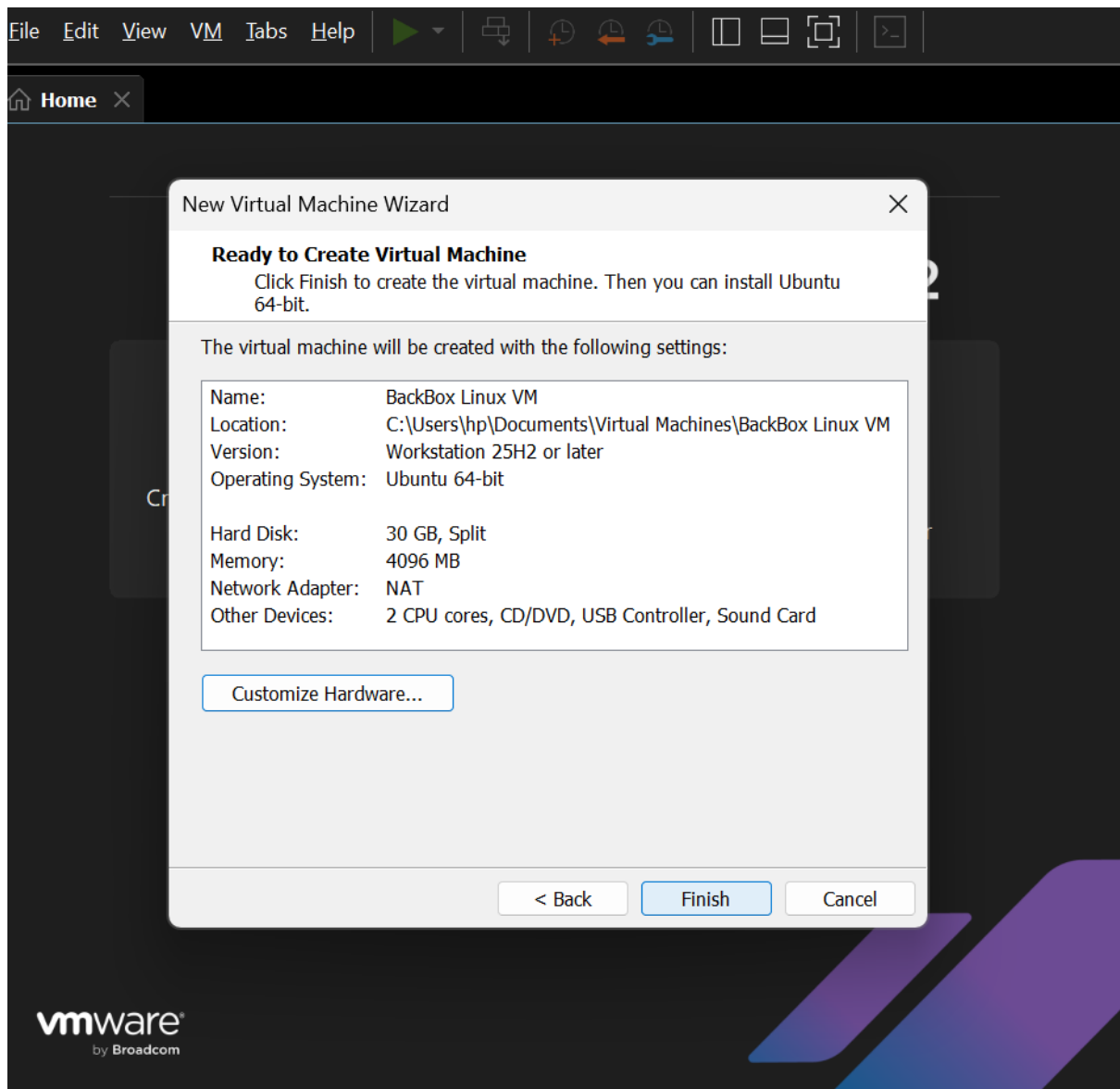
6. Hardware Customization

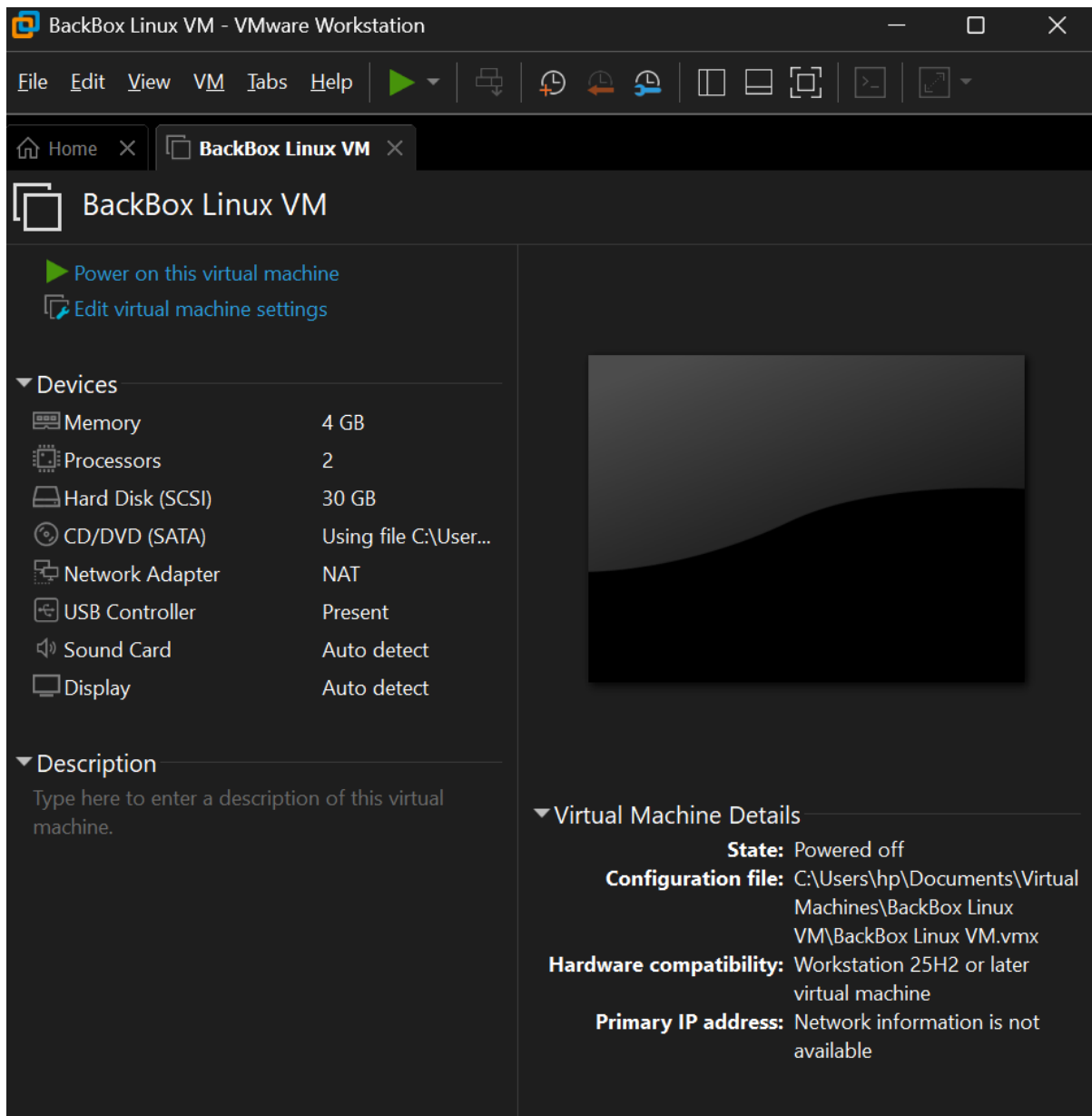
After clicking on “**Customize Hardware**”, the virtual machine settings were adjusted to optimize performance.

- Unnecessary devices such as **USB Controller** and **Sound Card** were removed.
- **3D Graphics Acceleration** was disabled to improve stability and avoid graphical issues during installation.
- After completing the configuration, the settings window was closed

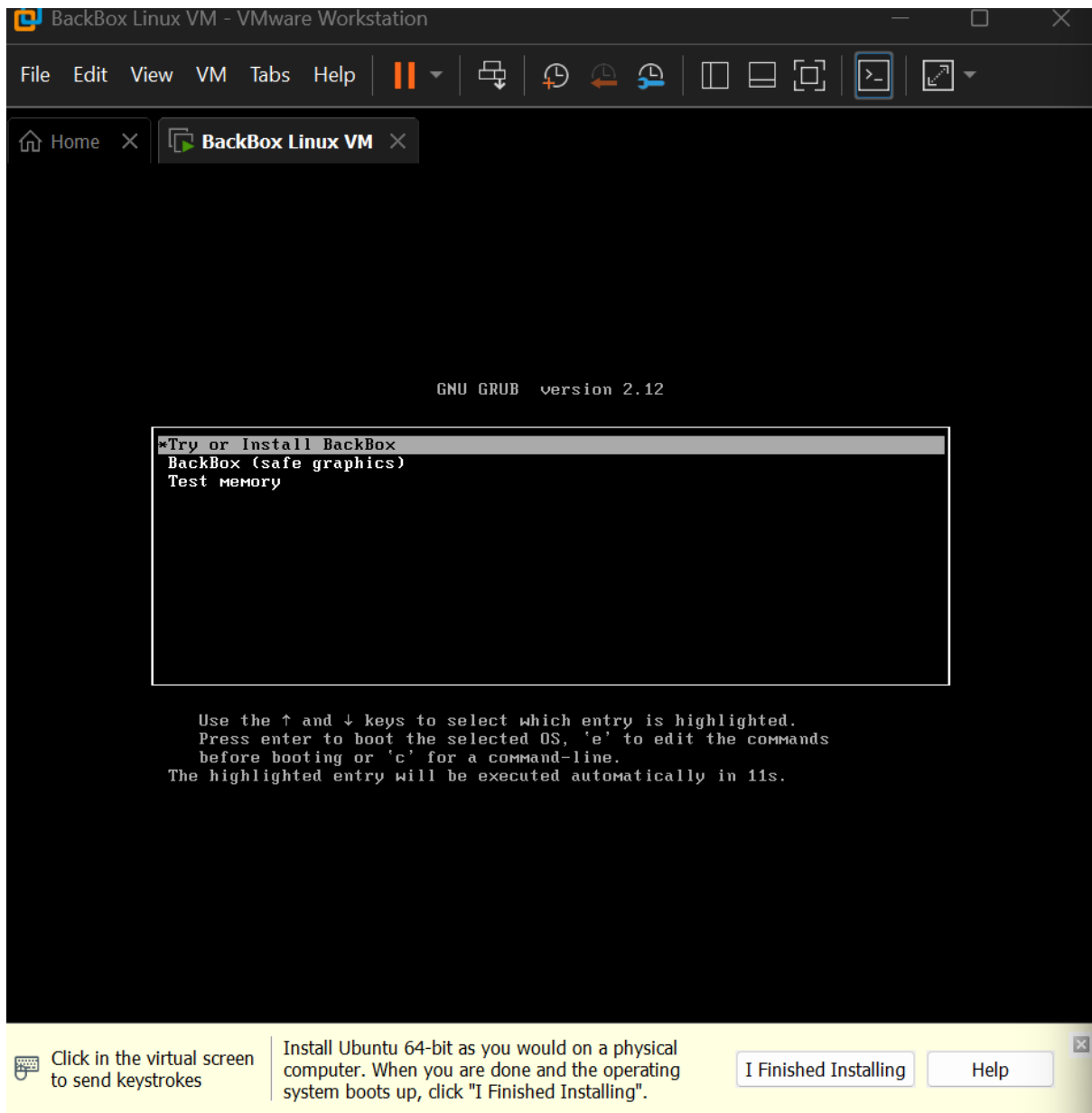


7. Review the configuration and click **Finish** to create the virtual machine.

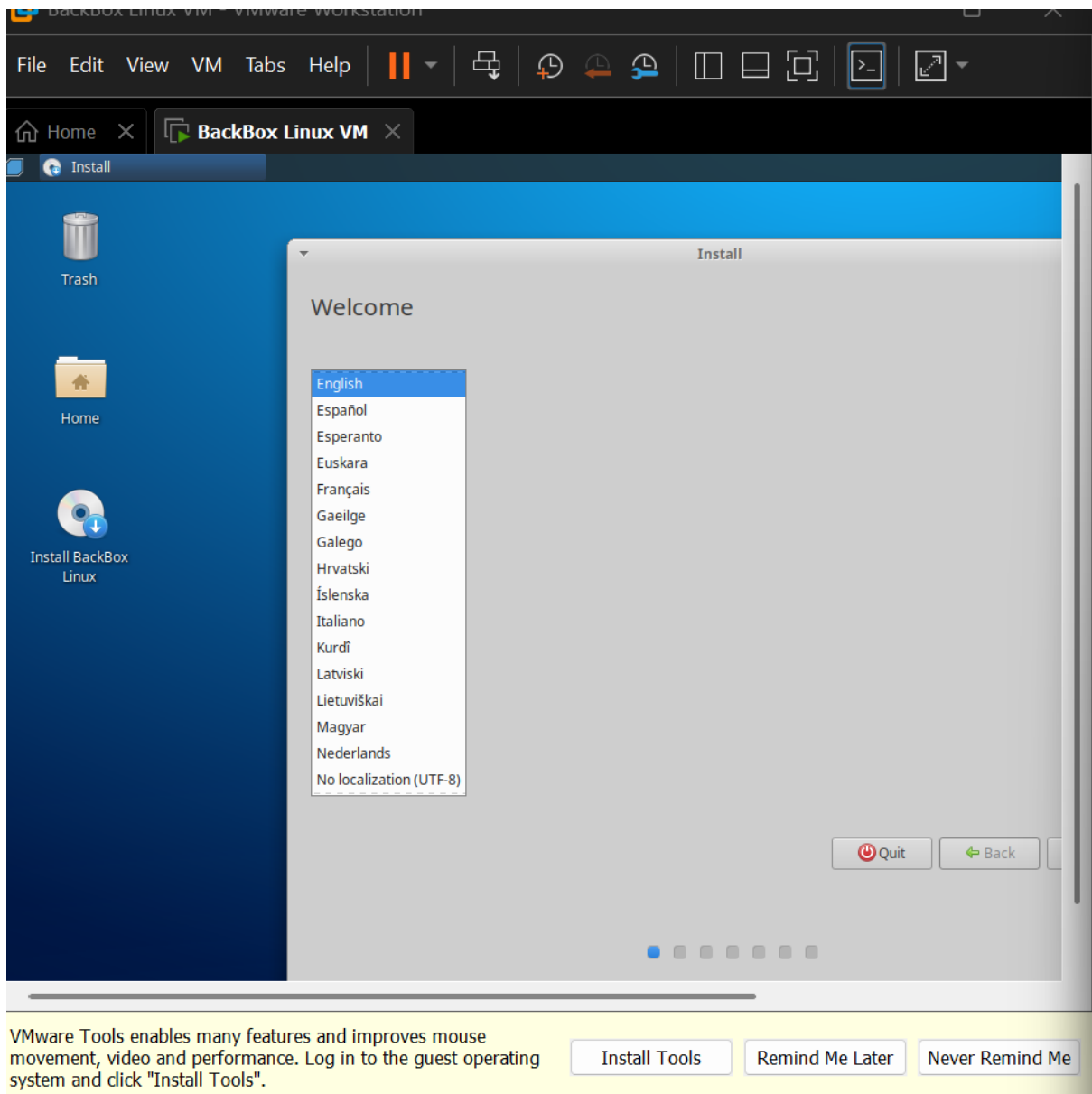


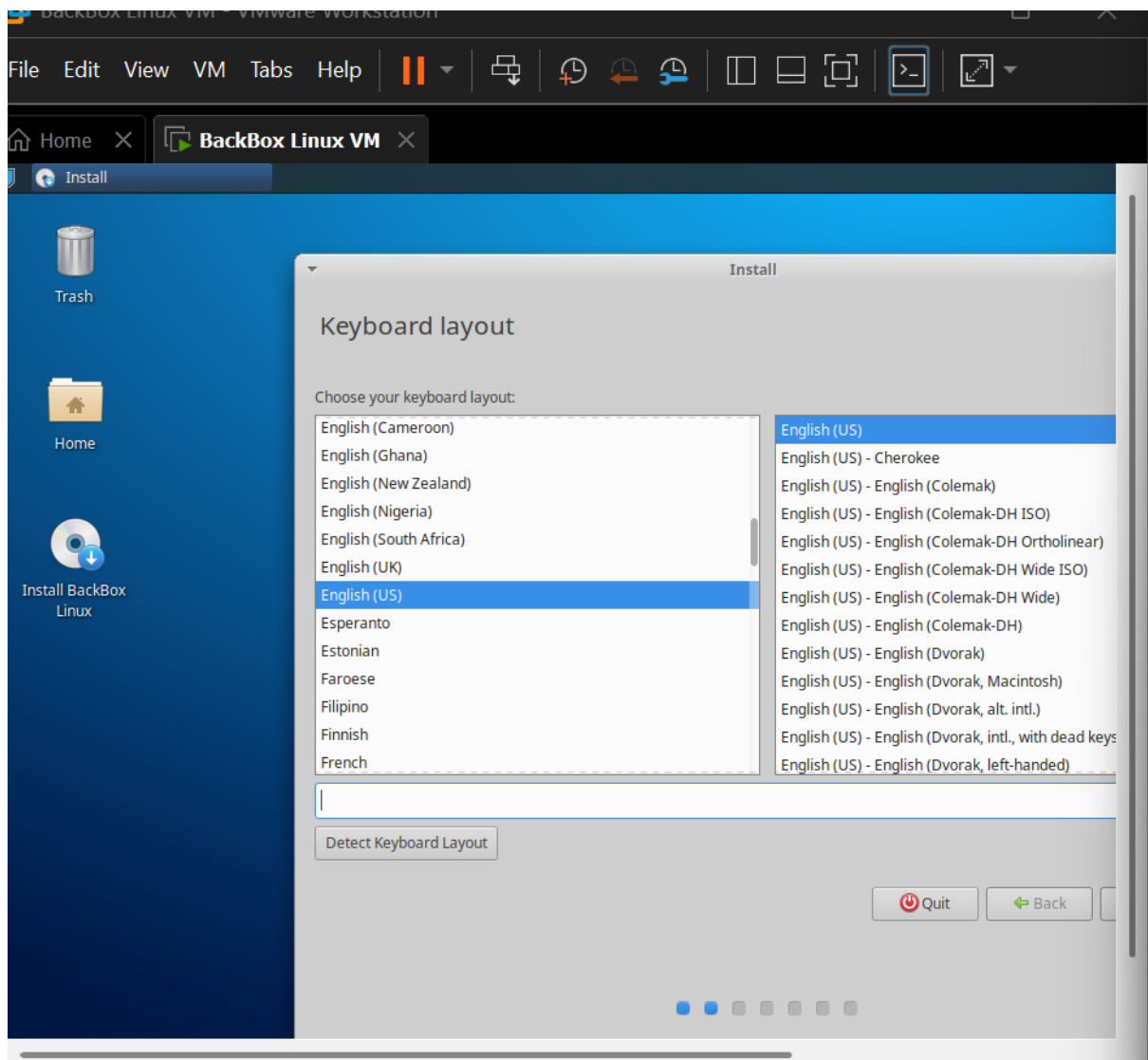


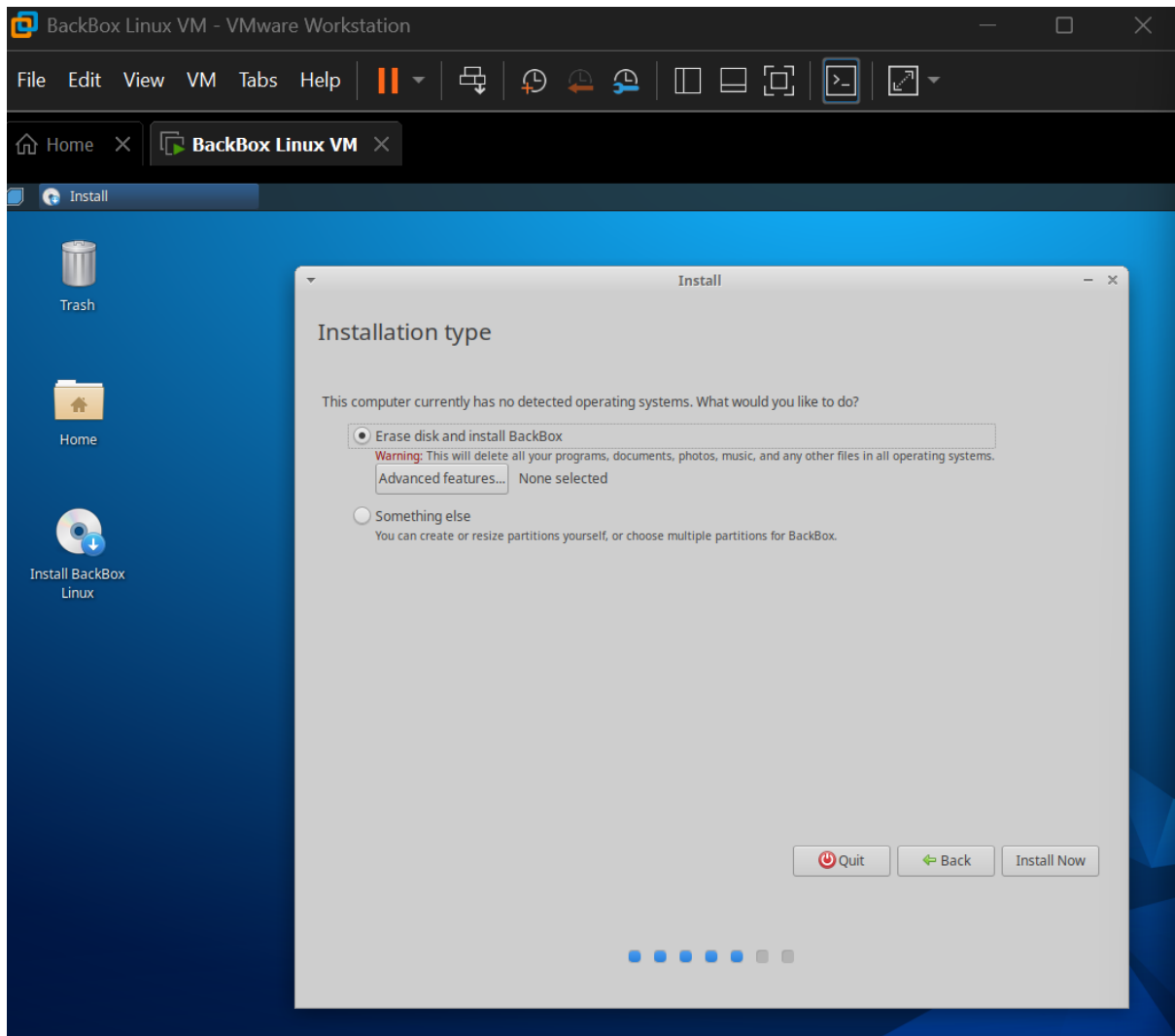
After completing the virtual machine setup, click the **green triangle (Power On)** button in VMware Workstation to start the system and boot the BackBox Linux ISO.

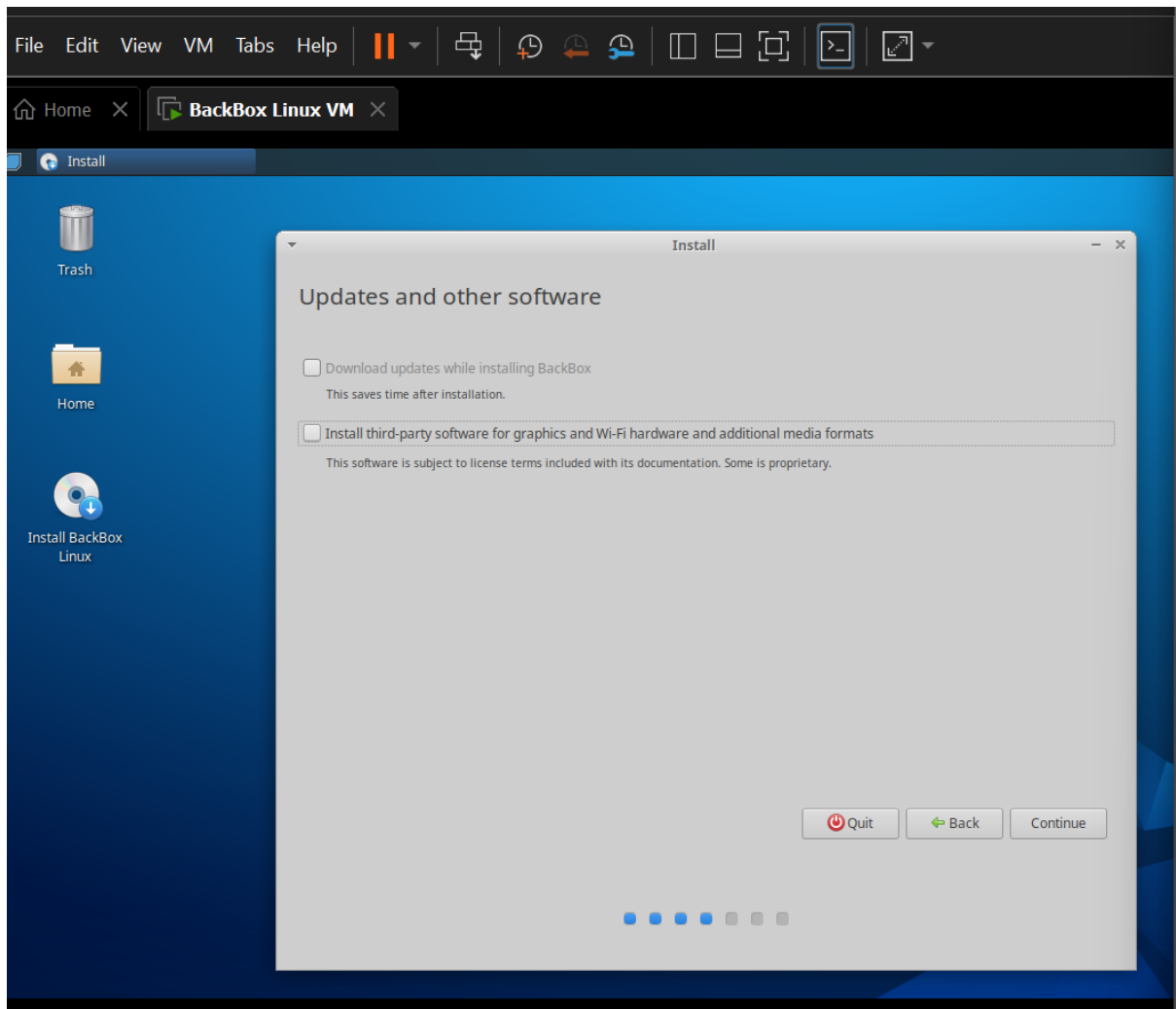


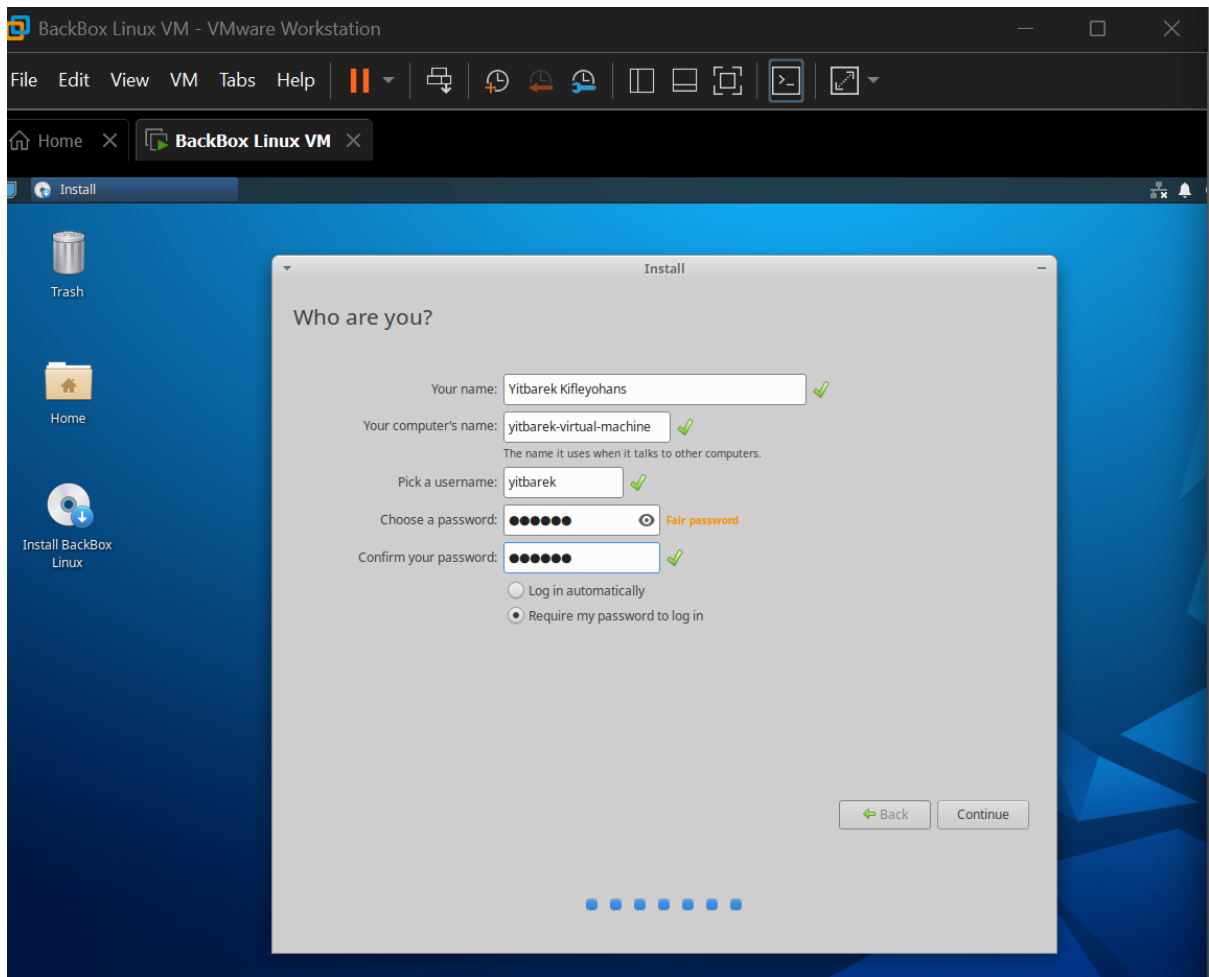
Complete language, keyboard, location, disk, and user setup.

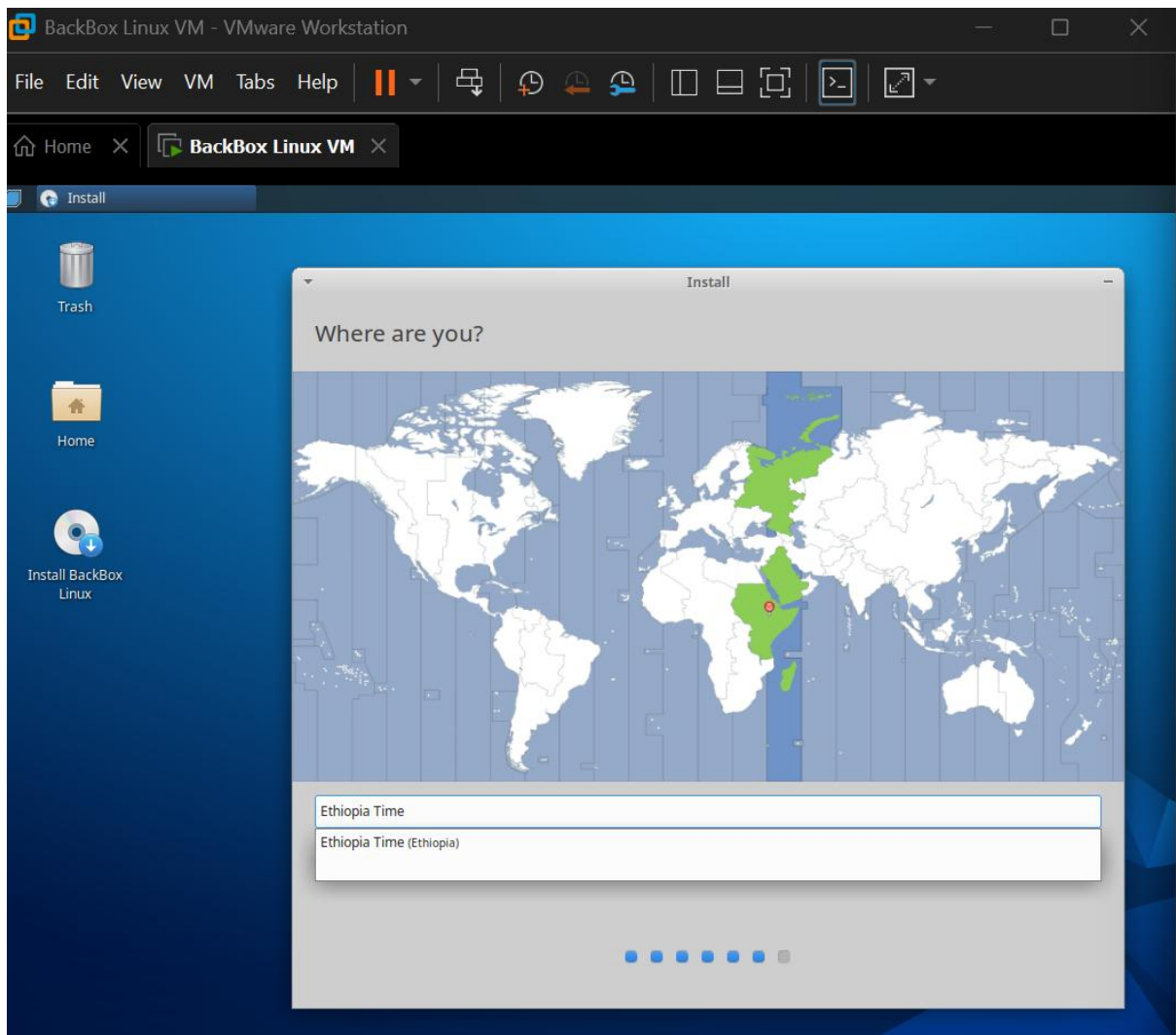




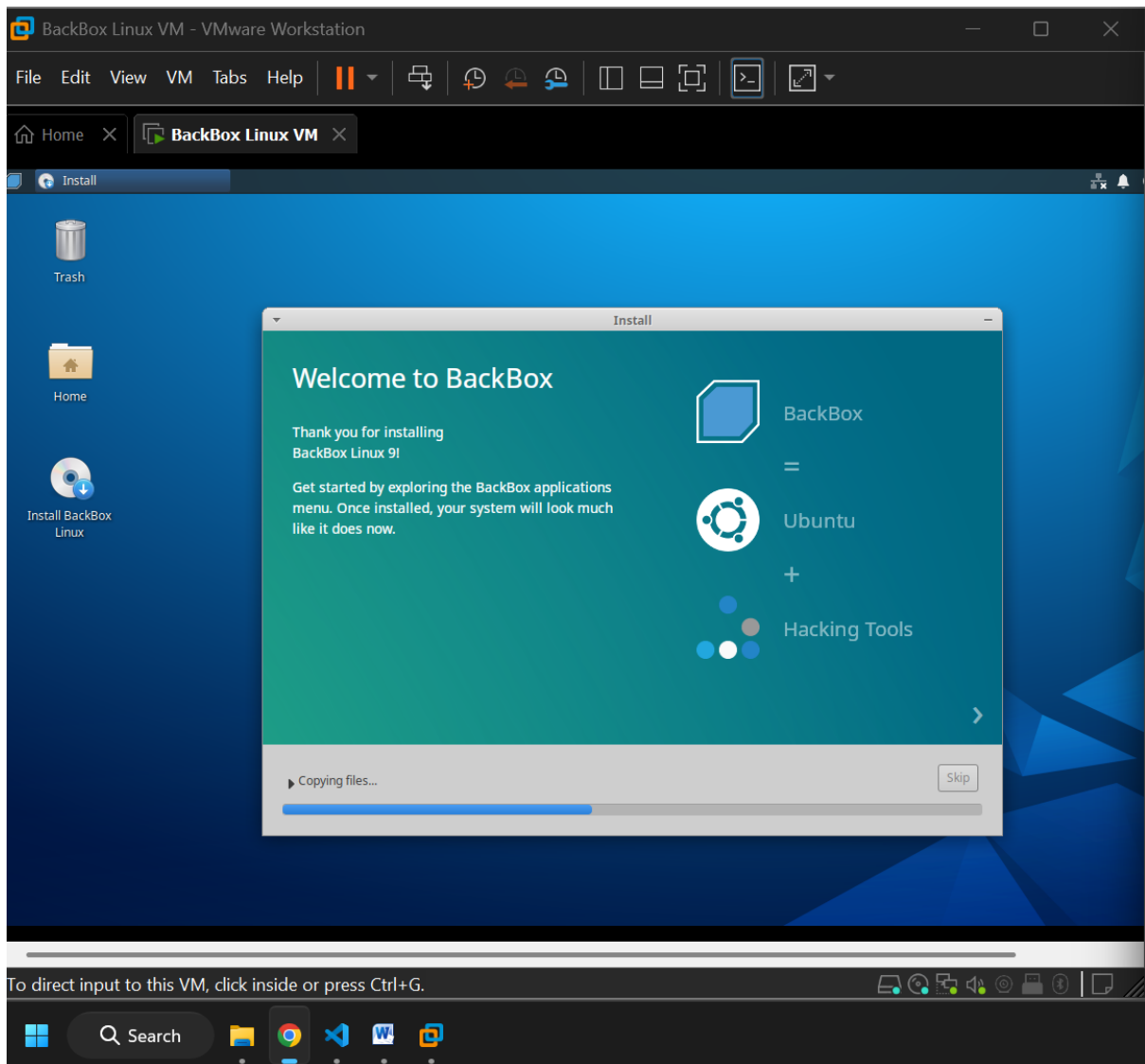




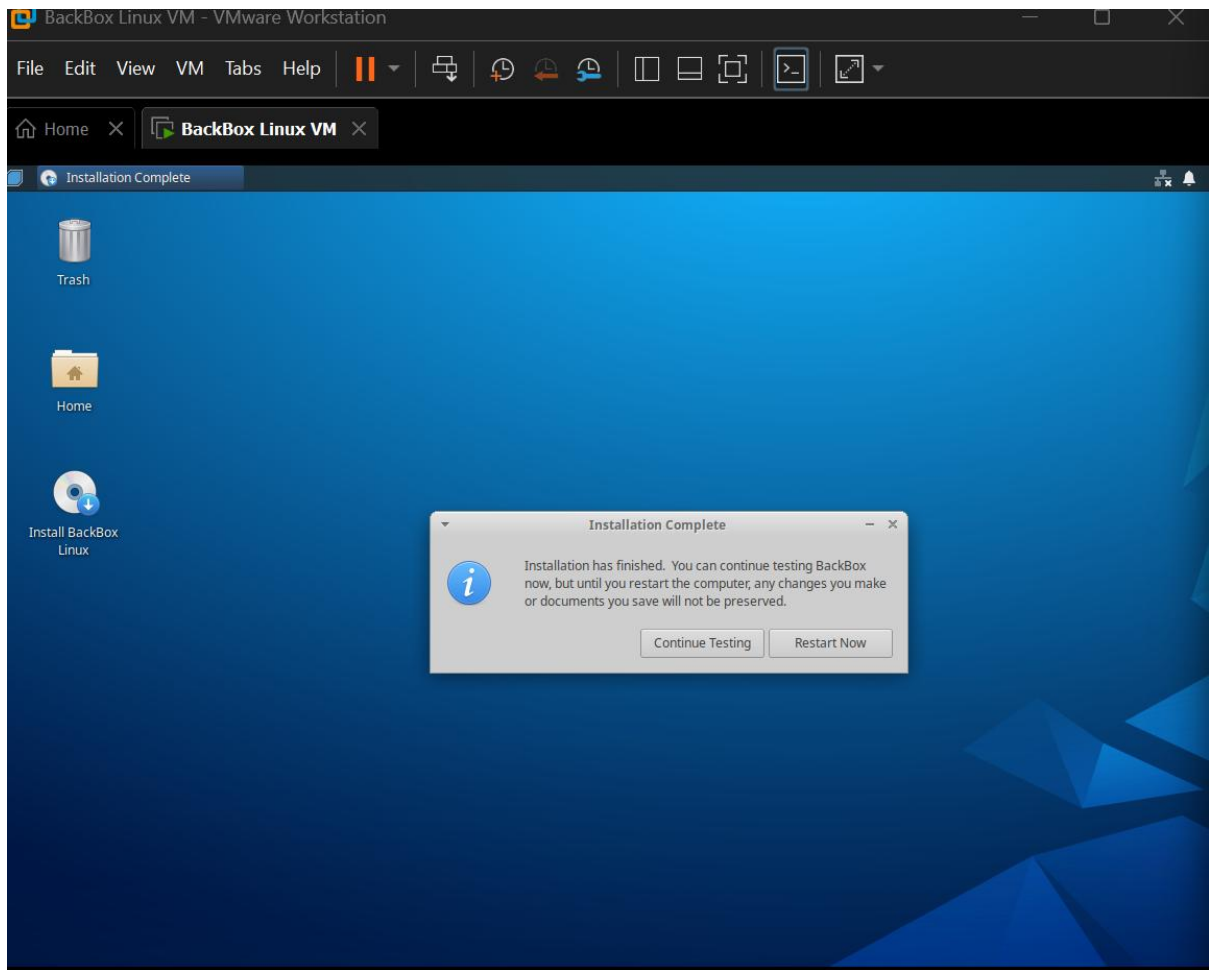




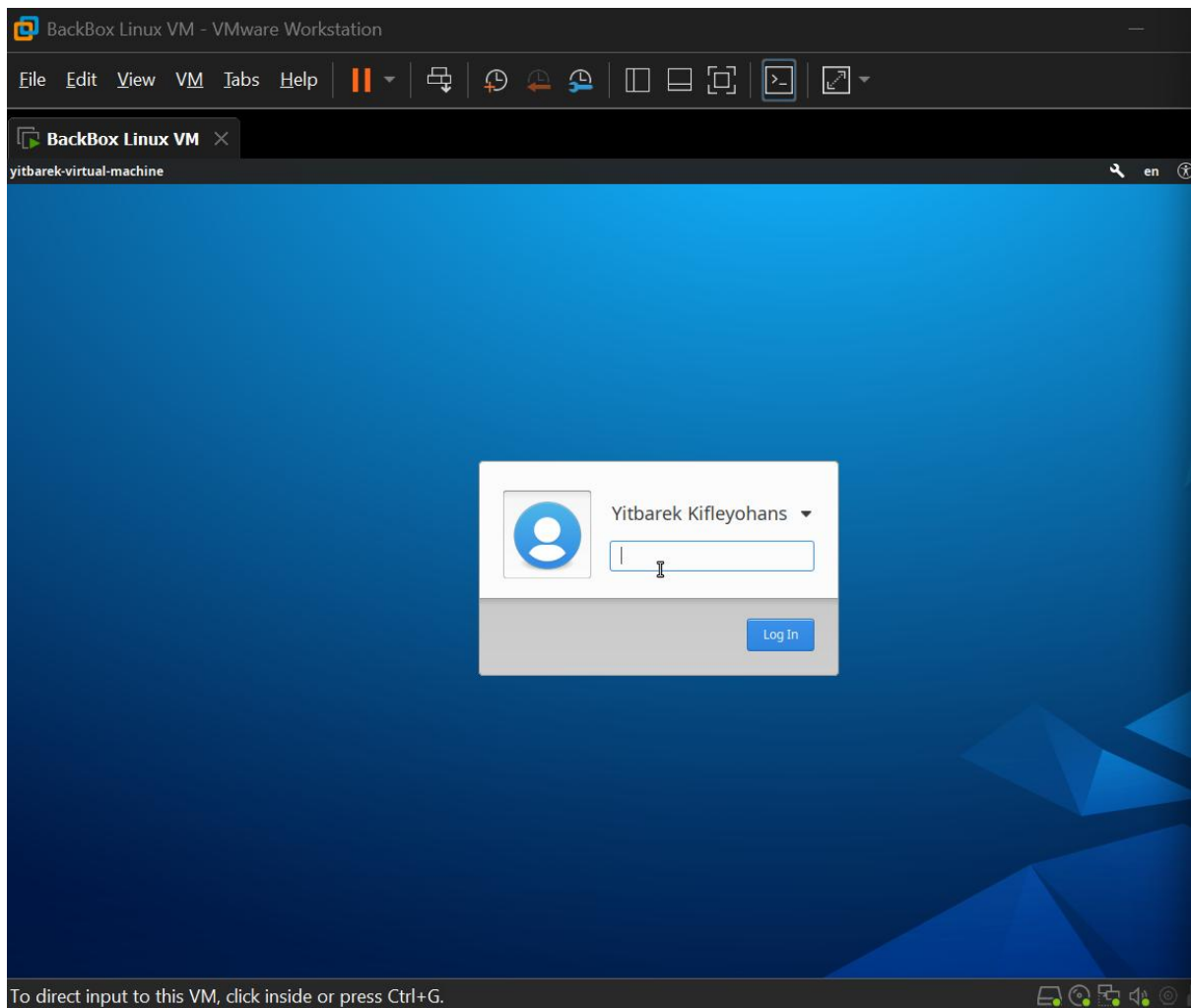
Wait for the installation to finish.



Click **Restart Now**



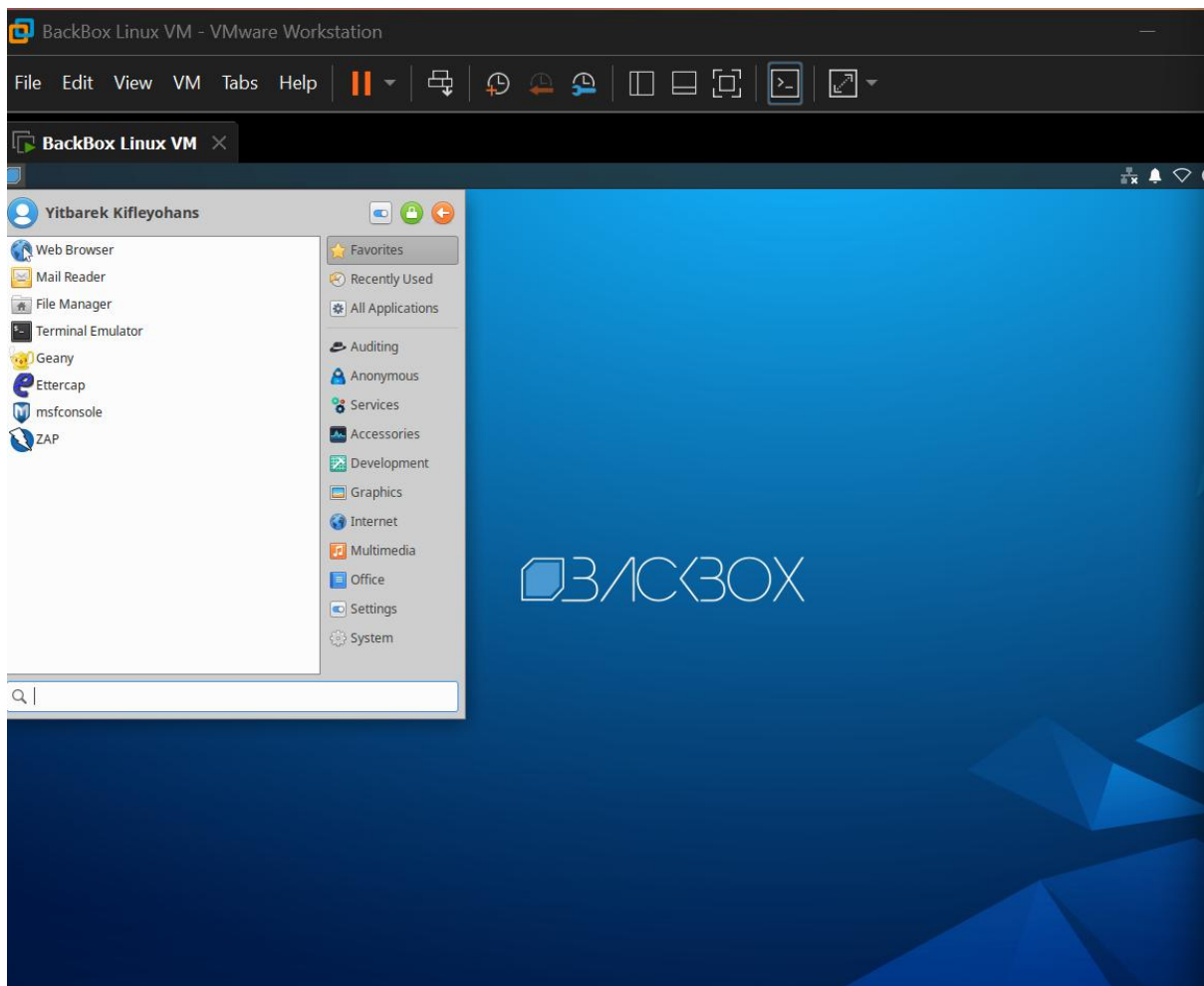
After reboot, log in to the system.

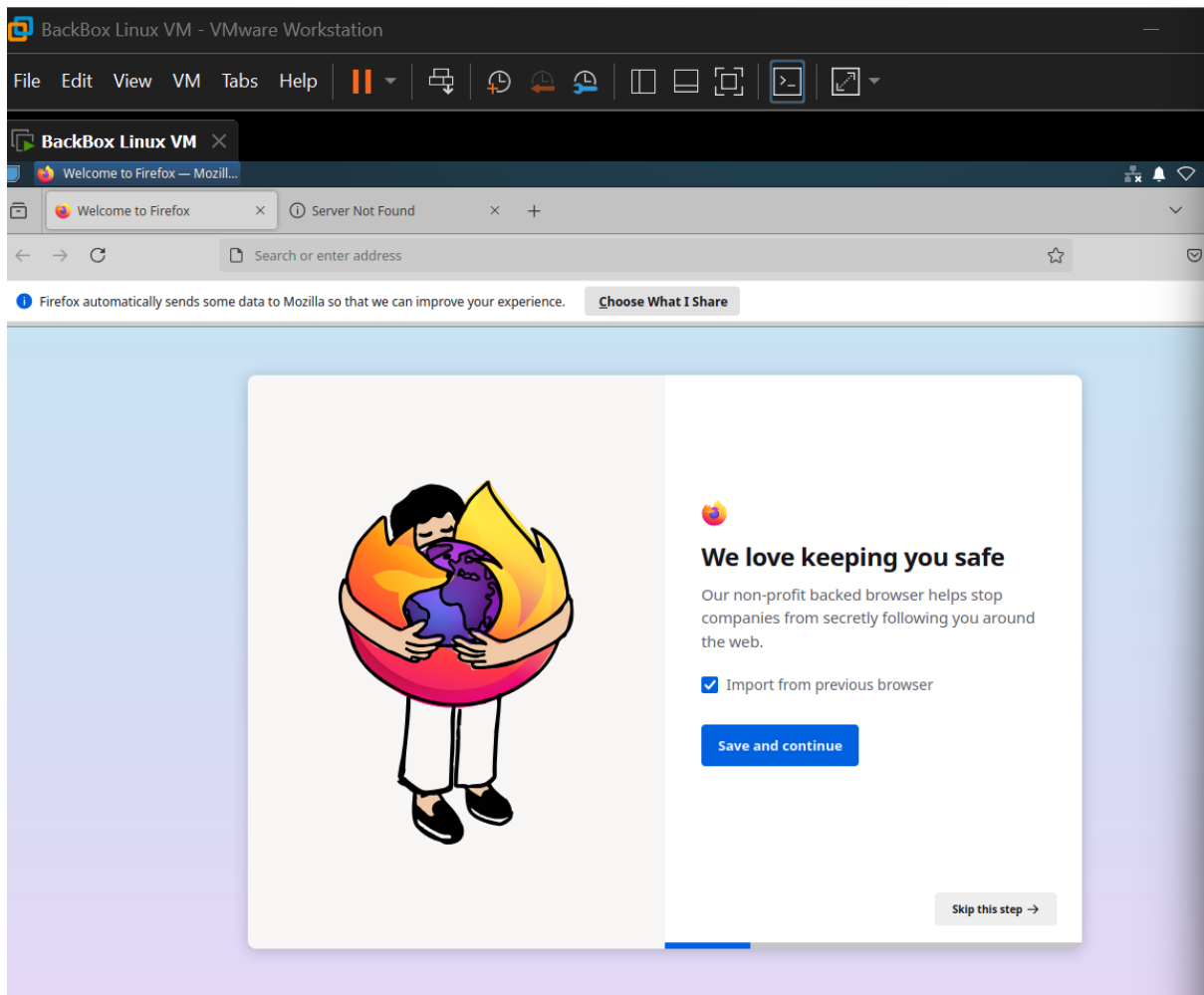


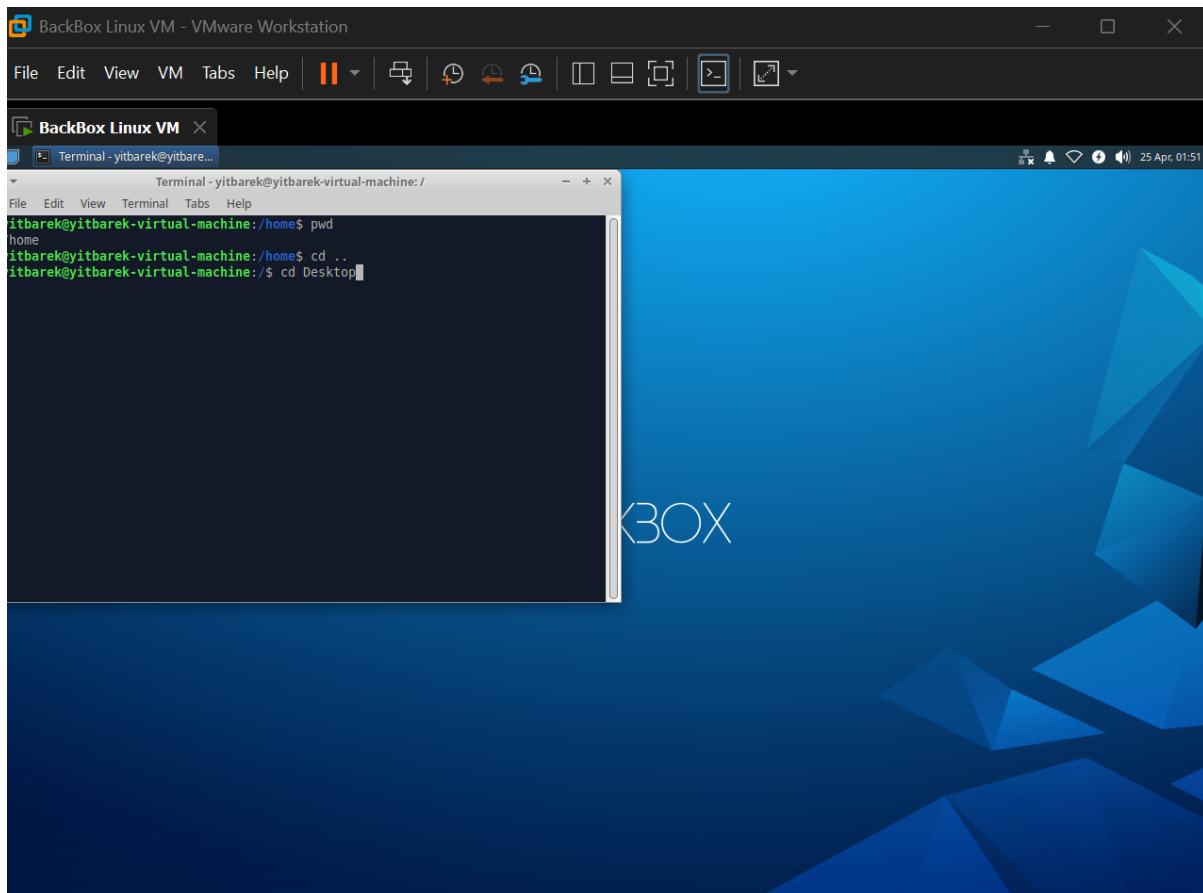
Post-Installation Verification

- Open the **Terminal** using **Ctrl + Alt + T** to test command-line functionality.
- Launch the **Firefox web browser** to verify internet access.
- Create a **new folder and file** to check file system operations.
- Confirm that the system is running smoothly without errors.

- Begin using BackBox Linux as a learning environment for security testing and system practice.







e. Issues

During the installation and setup process, a few minor issues were encountered. First, the correct BackBox file had to be identified carefully because the official download page offers both the real ISO image and a torrent option; only the `.iso` file is usable directly in VMware. BackBox Linux 9 is distributed as `backbox-9-desktop-amd64.iso` on the official download page, and the release notes indicate that it is a 64-bit image with a 30 GB disk-space recommendation and 1024 MB RAM minimum.

Another issue was the temporary unmounting warning that appeared at the end of installation. This was not a major failure, but it required attention before rebooting the virtual machine. In addition, the virtual machine had to be configured correctly as **Linux > Ubuntu 64-bit**, since BackBox is Ubuntu-based

f. Solutions

To solve the download problem, the correct **ISO file** was selected from the official BackBox download page instead of the torrent file. The virtual machine was then created in VMware Workstation using the Ubuntu 64-bit guest profile, which matches the Ubuntu-based design of BackBox Linux.

For the unmounting warning, the system was restarted normally and the ISO was detached from the virtual drive after installation. The virtual hardware was also simplified by removing unnecessary devices such as USB and sound, and 3D acceleration was disabled to improve stability during installation.

g. File System Support

i. Supported File System and Reason

BackBox Linux is an Ubuntu-based distribution, and therefore it primarily supports standard Linux file systems. In this project, the system was installed using the **ext4 (Fourth Extended File System)**, which is the default and most widely used file system for modern Linux distributions.

The ext4 file system was selected because of its **stability, performance, and reliability**. It supports large file sizes, efficient storage management, and improved journaling features that help prevent data corruption in case of system failure.

Compared to other file systems such as NTFS, FAT32, exFAT, Btrfs, ZFS, HFS+, and APFS, ext4 is more suitable for Linux environments because:

- NTFS and FAT32 are mainly designed for Windows systems
- exFAT is used for removable storage devices

- HFS+ and APFS are designed for macOS systems
- Btrfs and ZFS are advanced Linux file systems but are more complex and resource-heavy for basic virtual machine setups

Additionally, ext4 supports the standard Linux directory structure such as `/home`, `/etc`, `/bin`, and `/usr`, making it ideal for system organization and development environments.

h. Advantages and Disadvantages

Advantages

- BackBox Linux is lightweight and Ubuntu-based, which makes it easy to run in a virtual machine.
- It is suitable for penetration testing, security auditing, and other security-oriented tasks.
- Virtualization provides isolation, so the guest system can be tested without affecting the Windows 11 host.
- The VM can be restarted, copied, or deleted safely after testing.

Disadvantages

- The BackBox ISO is relatively large at 6.1 GiB, so the download takes time and requires patience.
- Running a VM uses real RAM, CPU, and storage from the host machine.
- If the virtual machine is not configured correctly, installation can show warnings or boot issues.
- Performance may be slower than running directly on physical hardware.

i. Conclusion

The installation of BackBox Linux on VMware Workstation was completed successfully in a virtual environment. The project demonstrated how virtualization can be used to install and test a Linux distribution safely on a Windows 11 host. BackBox Linux 9 was installed as a 64-bit Ubuntu-based system and configured for practical use in security-related learning and experimentation.

This project also showed the importance of selecting the correct ISO file, configuring the virtual hardware properly, and understanding filesystem support in Linux. The final result was a working BackBox Linux desktop environment with Terminal and Firefox available for verification and basic use.

j. Future Outlook

BackBox Linux can be used for more advanced practice in the future, including penetration testing labs, vulnerability assessment exercises, and network security experiments. Because the system is running inside a virtual machine, it can be safely extended with snapshots, additional tools, and isolated lab networks for deeper practice. VMware Workstation also supports desktop virtualization on Windows and Linux, making it practical for repeated experimentation and coursework.

2. Virtualization

What Is Virtualization?

Virtualization is a technology that allows the creation of multiple simulated environments, called **Virtual Machines (VMs)**, from a single physical hardware system. It uses software to abstract physical hardware components such as CPU, RAM, storage, and network resources so they can be divided and allocated to different virtual systems.

Instead of a one-to-one relationship between hardware and operating system, virtualization enables a **one-to-many relationship**, where multiple operating systems can run independently on the same physical machine.

Key Concept: Virtualization creates software-based representations of computing resources such as servers, storage, networks, and applications.

Why Use Virtualization?

Virtualization provides several important advantages:

- **Cost Efficiency:** Reduces the need for multiple physical machines, lowering hardware and energy costs.
- **Resource Optimization:** Improves utilization of system resources, which are often underused in physical servers.
- **Isolation and Security:** Each virtual machine operates independently; failure or malware in one VM does not affect others. This makes it ideal for penetration testing environments such as BackBox Linux.
- **Snapshots and Backup:** Virtual machines can be saved as snapshots, allowing restoration to previous states when needed.

- **Portability:** Virtual machines can be moved between systems without reconfiguration.
-

How Does Virtualization Work?

Virtualization operates through an abstraction layer that separates physical hardware from virtual machines.

Main Components:

- **Host Machine:** The physical computer that provides hardware resources such as CPU, RAM, and storage.
- **Hypervisor:** The software layer that manages and allocates resources to virtual machines.
- **Guest OS:** The operating system installed inside the virtual machine (e.g., BackBox Linux).
- **Virtual Resources:** Simulated hardware components such as virtual CPU, RAM, and network adapters.

The hypervisor ensures that each virtual machine operates independently without interfering with others.

. References

- BackBox Linux Official Documentation.
<https://www.backbox.org>

- VMware Workstation Documentation.
<https://www.vmware.com>
- Oracle VirtualBox User Manual.
<https://www.virtualbox.org>
- Ubuntu Official Documentation (Base System Reference). <https://ubuntu.com>
- Linux Filesystem Hierarchy Standard (FHS).
https://refspecs.linuxfoundation.org/FHS_3.0/fhs/index.html
- Wikipedia Contributors. “Virtualization.”
<https://en.wikipedia.org/wiki/Virtualization>
- GitHub Community Projects related to Linux virtualization and penetration testing environments.

Acknowledgment

This project is submitted for review by **Mr. Wondmu Baye**, OSSP Lecturer at **Bahir Dar University (BDU)**.

Thank you for taking the time to review this work. Your feedback is highly appreciated.

If you find this project useful, feel free to **star** the repository and **fork** it so that others can learn from and build upon it.

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I sincerely appreciate everyone who reviewed this project.

Yitbarek.k